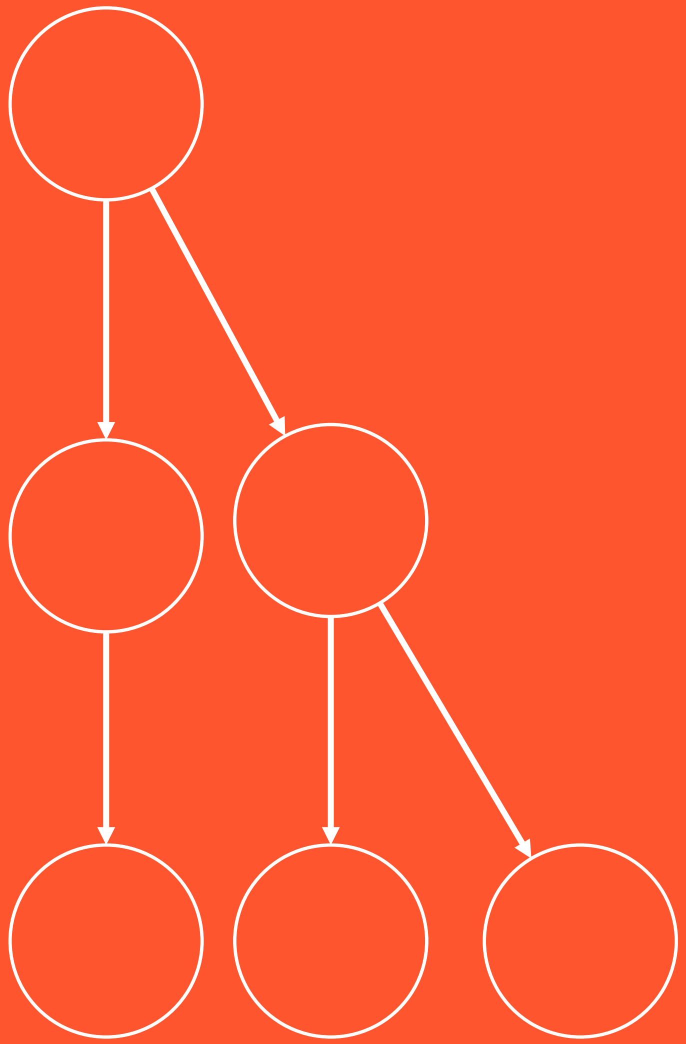


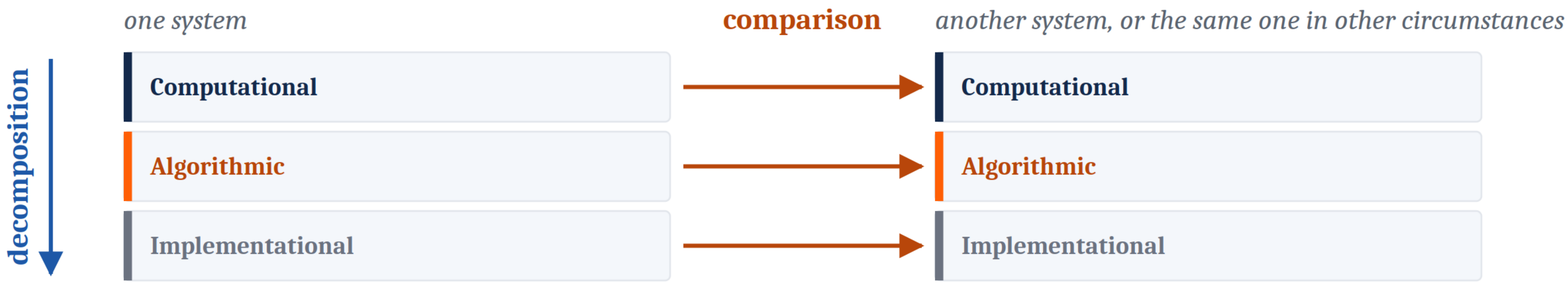
BCOG 100

Comparative Approaches
Lecture 2.1 — The Comparative Method

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

A Second Method, at Right Angles to the First

Two methods, at right angles.



Marr goes down. Comparison goes across. A field with both can ask what a capacity is, and what else in the world has anything like it.

A field with both can ask what a capacity **is**, and also ask **what else in the world has anything like it**.

The practice has a name: the **comparative method**.

One Figure, Two Methods

Down a column is **Marr**. Across a row is **comparison**.

The frog is **not** a poor version of a human visual system. It is solving a different problem, and its machinery fits that problem well.

	Human vision	Frog vision
Computational *What problem is being solved?*	Recover a general description of surfaces and objects — one that can serve many tasks at once	Find a small dark thing moving as an insect moves, and strike; detect a spreading shadow, and flee
Algorithmic *By what procedure?*	Build the description in stages, from local edges toward viewpoint-stable objects	Compute a few specific features and route them almost directly to action
Implementational *In what physical stuff?*	Retina, a thalamic relay, and a large cortical hierarchy	Much of the work is done in the retina itself; output goes chiefly to the optic tectum in the midbrain

What Would You Conclude?

A person and an image-recognition model both label the wolf as a dog.

Across a large test set:

- the person is right about **95%** of the time
- the model is right about **95%** of the time
- the two **agree with each other 95%** of the time

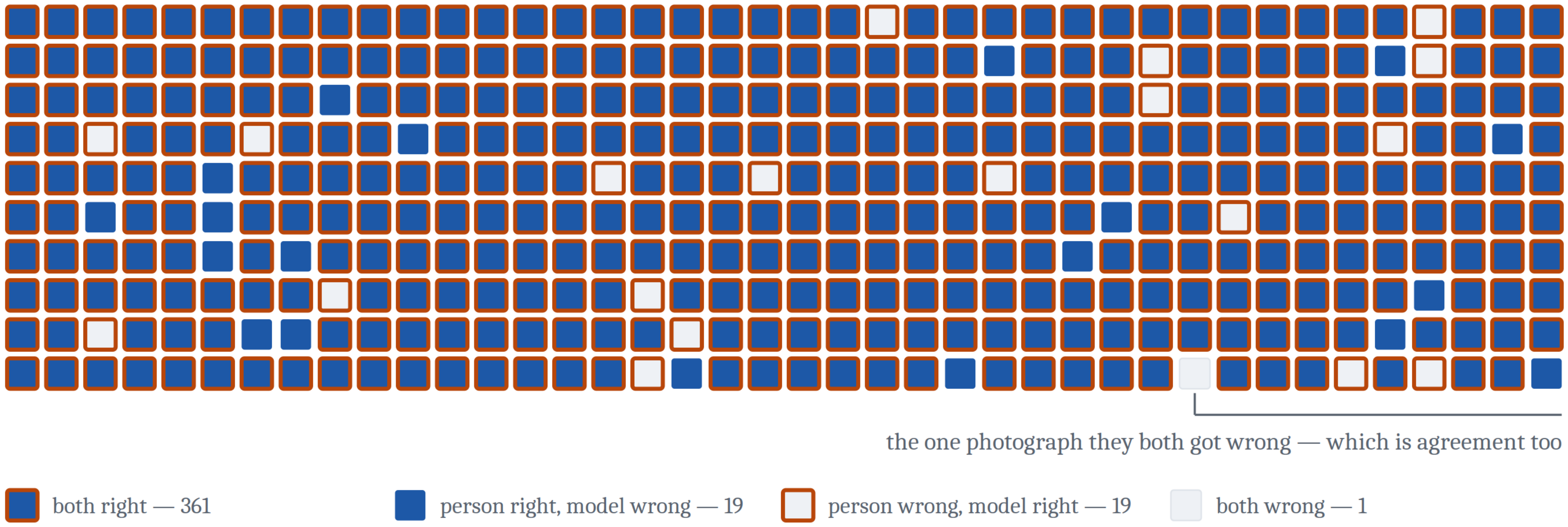
This is the kind of finding reported as evidence that the model sees the way people do.

Does it show that?

A Comparison Cannot Interpret Itself

Fill = the person got it right. Thick border = the model got it right.

Each marks 95% of the same 400 photographs, and the two have nothing to do with each other.



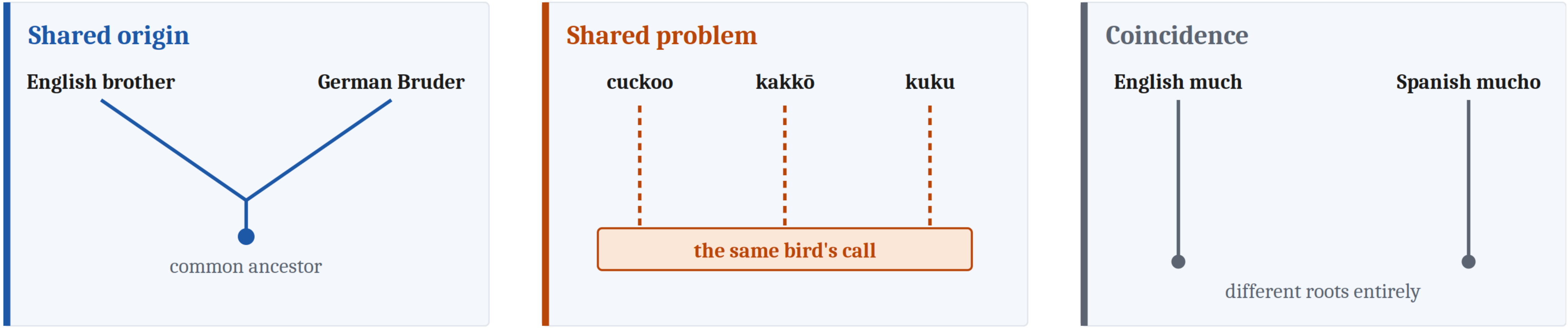
362 of the 400 carry the same verdict — 90.5% agreement, between two systems that have never met.

An observed 95% is 4.5 points more than that. The surplus is the finding.

Two **independent** systems, each right 95% of the time, with random errors, will still agree about 90% of the time.
But 95% agreement is about five points above what independence alone produces.
It is this **surplus** that calls for an explanation, not the agreement.
Noticing that two things are similar is the easy part. The work is figuring out *why*.

Question One: Where Did the Similarity Come From?

Where did the similarity come from?



inherited from one ancestor language

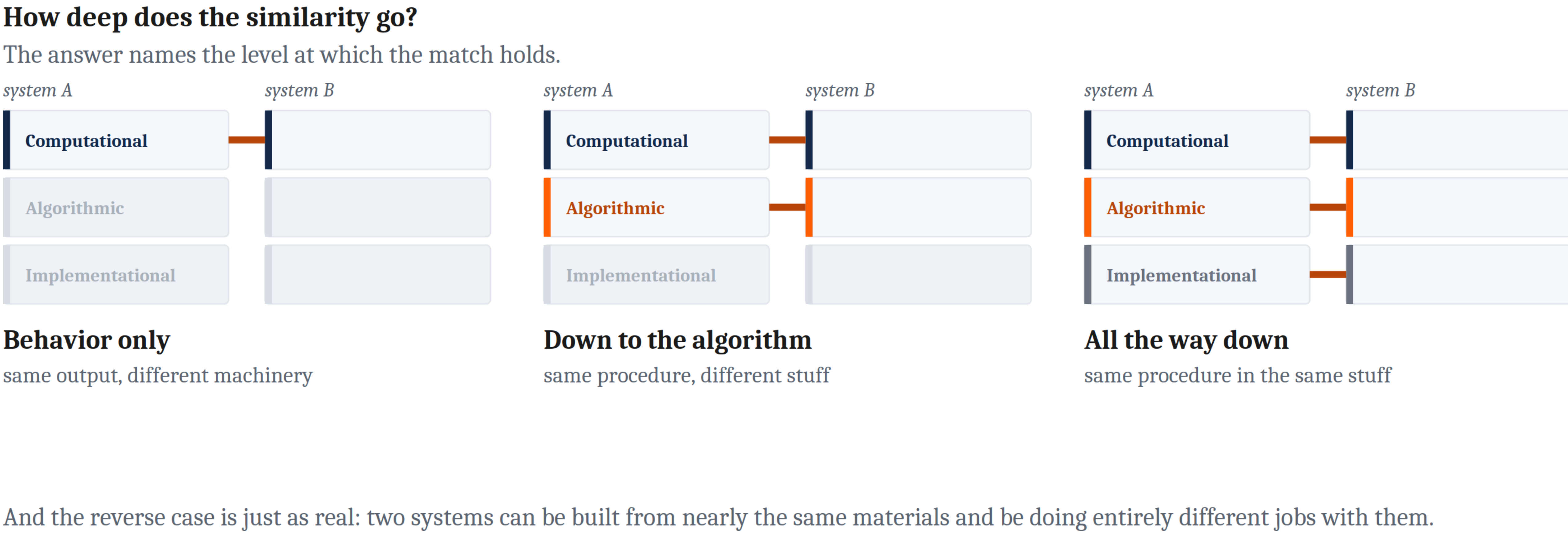
English, Japanese, Hindi — unrelated, one bird

mycel and multus, no connection

Only the first lets us reason about an ancestor. The second says there was none — which is a finding, not a consolation prize.

Shared origin lets us reason about the ancestor. Shared problem licenses **no** claim about an ancestor at all — arriving twice independently *is* the claim that the ancestor lacked it. Coincidence teaches nothing.

Question Two: How Deep Does It Go?



A merely behavioral match teaches less than we might think.

And the reverse case is just as real: two systems can be built from nearly the same materials and be doing **entirely different jobs** with them.

Question Three: How Precisely Can We State It?

How precisely can we state it?

One behavioral claim, sharpened. Each step is the same finding, looked at harder.

both tell the two categories apart	both do it at about the same rate	both miss the same kinds of item	same errors, same items, same proportions

loose → precise

Precision is not yes or no. It is how finely we looked before we stopped looking — and looking harder tends to surface differences.

Depth asks **at which level**. Precision asks **how finely specified** the match is at whatever level we are looking — so a claim can be **shallow and precise**, or **deep and vague**.

To say two systems behave alike is never to say they would behave alike in **every** circumstance. It is to say they matched on what we happened to measure.

Three Questions, Not Three Steps

one observed resemblance

Origin
where did it come from?

— shared origin
— shared problem
— coincidence

Depth
what level does it hold at?

— behavior only
— down to the algorithm
— all the way down

Precision
how finely can we state it?

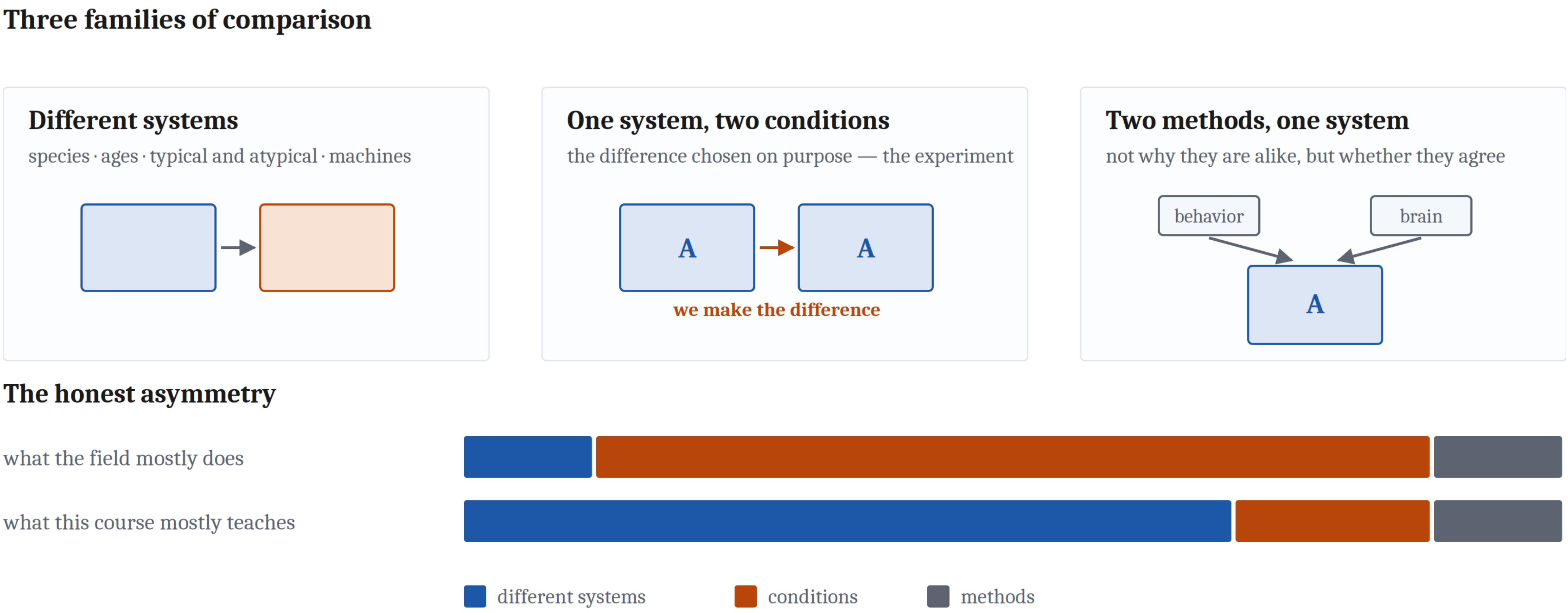
— loose — on average
— sharper — the same items
— precise — the same proportions

Three questions, not three steps. They cross, they do not nest — and a shared mechanism is itself a resemblance.

They **cross**. They do not nest, and they are not a sequence.

And a shared mechanism is **itself a resemblance** — which sends us straight back to question one.

Three Families of Comparison



The same three questions run through all three families. Only the vocabulary changes.

Four Ways This Goes Wrong

Anthropomorphism — projecting human qualities onto a system that has not earned the description. The dog that looks guilty. The chatbot that sounds like it means it.

Anthropocentrism — treating the human case as the reference point, the default, the "normal" one.

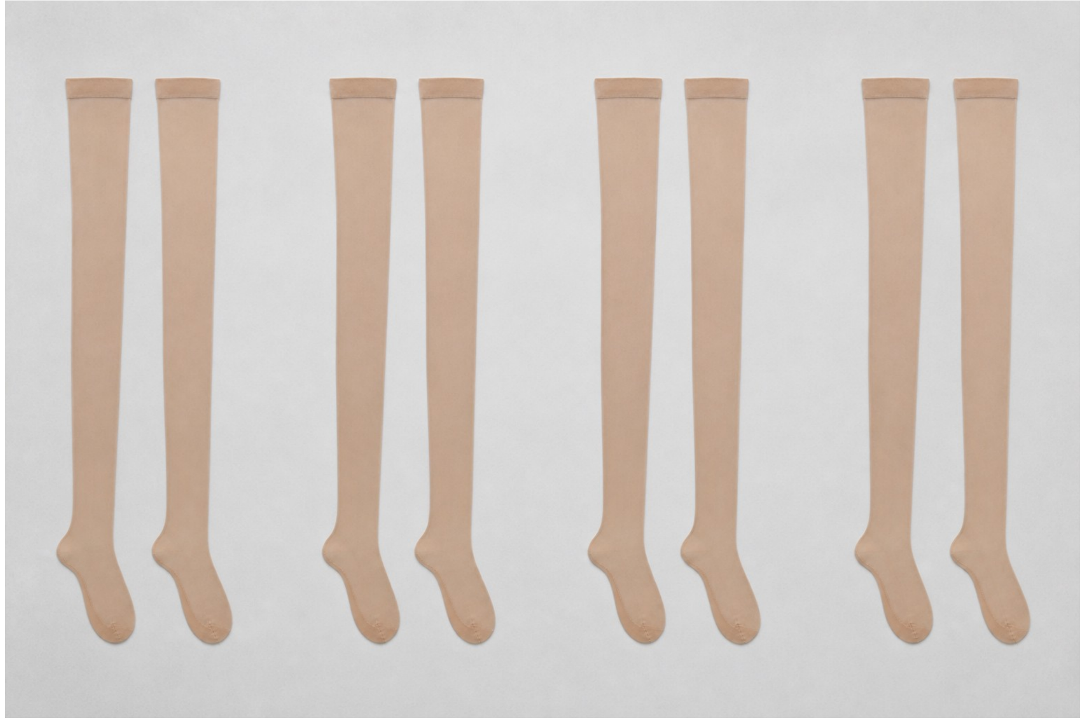
Unconstrained adaptationism — assuming every trait must be a finely tuned solution to some problem.

Four Ways This Goes Wrong

An error about depth and precision — and the one underneath the other three.

Surface matching is the theme under the other three. It is an error about **depth** and **precision**: two systems that behave alike get treated as alike underneath, when the only things checked were the handful of dimensions somebody happened to measure.

And We Do It to Ourselves



Nisbett and Wilson. Shoppers were shown several pairs of stockings laid out in a row.

The pairs were **identical**.

Shoppers chose the **rightmost** pair far more often than any other.

Asked why, they explained their choice by the **knit and the sheerness of the fabric** — and denied that position had anything to do with it.

We often have no direct access to why we did something. We generate a plausible reason instead.

None of This Is Neutral

Arguments about whether animals think, whether a fetus has experiences, whether a machine understands anything, are very often powered by something other than the evidence.

Claims about **who has which capacity** have historically tracked claims about **who is owed what**.

The pull runs both ways. Wanting to raise an animal's standing is a reason to read its behavior generously. Not wanting to change how animals are used is a reason to read the same behavior thinly.

Neither motive makes anyone wrong — but motives shape which studies get run, which results feel convincing, and which get explained away.

Also in the chapter, and still fair game: comparison across conditions (the experiment, confounds, control conditions) · across ages · across typical and atypical minds.

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3	Marr's three questions, asked of human and frog vision	fig-levels-vision.png	CC BY 4.0	Jon Willits — drawn for this lecture from the Module 2 reading's table · own work
4	Two wolves	Black_and_White_Wolves.jpg	CC BY 2.0	Stéfan, via Wikimedia Commons · https://commons.wikimedia.org/wiki/File:Black_and_White_Wolves.jpg
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7	A match that stops at the top, in the middle, and all the way down	fig-depth-levels.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
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